Presentation prep — read this before you go up

Score: 59.7 / 100. D1 12.1/25 · D2 29.8/35 (**strongest tier**) · D3 14.2+/40 Started the day at 0. Was 49.9 an hour before the end.

1. What we built — say this first

> **A windowed vision-language pipeline that watches long video and reports anomalies with timestamps — plus a frozen-encoder classifier that catches what the VLM is blind to.**

```
video ──► sample frames BY TIMESTAMP ──► 20s sliding windows
                    │                              │
                    ▼                              ▼
      SigLIP2 + trained GRU head      Qwen3-VL-8B (prompted)
      "what is this, over time"       "what is happening here"
                    │                              │
                    └──────────► merge ◄───────────┘
                                   │
                        per-difficulty policy
                                   │
                          submission JSON
```

The one-sentence version of why it works: The VLM finds *appearance* — fire, smoke, flood, crashes. The frozen encoder finds *duration* — loitering, stalled vehicles. Neither does both.

2. The three-agent system — how we actually worked

Three Claude Code sessions, on **one repository**, talking to each other by direct message.

Agent	Role	Owned
Agent 1 (coordinator)	Research, strategy, all GPU runs, all commits	`src/vad/`, `modal_app.py`, `status/`
Agent 2 (builder)	Data pipeline, synthetic video generation, training, scoring code	`src/ahc_vad/`, `scripts/`, `tests/`
Agent 3 (comms)	Demo, slides, process doc	`demo/`, `slides/`, `docs/`

The rule that made it work: ownership by directory. No agent edits another's files. Cross-boundary changes are *requested*, not made.

We learned that the hard way. Early on, Agent 1 and Agent 2 both built the same four modules — scoring, submission, event types. We threw away Agent 1's copies. After that, one rule fixed it.

One committer. Agent 1 reviewed and pushed everything. The repo is a graded deliverable; three agents committing in parallel would have made the history unreadable.

They corrected each other, out loud. Real examples worth quoting: - Agent 2 caught Agent 1 using **first-fit instead of best-IoU** matching in the scorer — silently wrong whenever predictions overlap. - Agent 3 caught a **Gemini run that exited "successfully"** having done 6 of 34 videos, writing the 28 failures to disk as confident "normal" predictions. - Agent 1 published *"bigger is worse, 4B beats 8B"*, then

retracted it when asked for the mechanism — the models differ on 3 videos out of 24, p = 0.25. **That retraction was load-bearing: the 8B produced our best score.**

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3. The knowledge factory — how three agents stayed coherent

Nothing important lived in chat. Everything went into files.

File	What it is
`.context/`	**Shared memory.** Problem, dataset, prior art, decisions. Every finding written down, with provenance: *measured*, *read in a paper*, or *assumed*.
`status/ledger.md`	**The experiment ledger.** Every technique across 7 tiers — input, prompting, decoding, frozen heads, fine-tuning, RL, cascades — marked ✅ done / 🔄 running / ⬜ planned / ⛔ rejected **with the reason**.
`status/models.md`	Why each model was chosen, what it was compared against, what we rejected before testing.
`status/experiments.md`	Every run and its verdict, including the failures.
`plans/`	Specs written before code.

Why it mattered: an agent could crash, restart, and pick up from the files. That happened. Twice.

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4. Models we tested — the headline table

All five on an identical 24-video set, same prompt, same policy, Modal A100.

Model	Size	Precision	Recall	F1	Verdict
Qwen3-VL-8B-Instruct	8B	0.62	0.40	0.48	**Shipped — produced our D1 + D2**
Qwen3-VL-4B-Instruct	4B	0.77	0.50	0.61	Ties the 8B (differ on 3 of 24)
Cosmos-Reason2-8B	8B	0.57	0.20	0.30	❌ Half the recall
Qwen3-VL-2B-Instruct	2B	0.30	0.15	0.20	❌ Too small
Qwen2.5-VL-7B-Instruct	7B	0.00	0.00	0.00	❌ Silent on 20 of 24
SigLIP2 + GRU head	frozen	—	—	—	**Shipped — +2.4 marks**

Three findings worth saying out loud:

1. **Benchmark leadership does not transfer.** Cosmos-Reason2-8B **tops NVIDIA's own Traffic Anomaly Reasoning leaderboard** and got *half the recall of a 4B* on our task.
2. **Qwen2.5-VL-7B is ASK-HINT's own published backbone** (89.83% AUC on UCF-Crime in the paper). Here it returned a bare `[]` on 20 of 24 videos. Instruction-following failure, not blindness.
3. **We nearly didn't test SigLIP.** It was marked ⛔ *rejected* in our ledger on the theory that a frozen probe can't use context. It turned out to be one of the most effective approaches on this task. **The evidence was there; our theory was wrong.**

Also tried and rejected before testing: hosted models (banned from the runtime path), 72B/90B (too large), Llama-3.2-Vision (accepts **one image per request**; we send 8–32 frames).

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5. Platforms

Platform	Used for	Outcome
Modal	Every GPU job — A100-40GB and A100-80GB	✅ The workhorse. ~$12 of $30 credit.
Google Gemini API	Reference ceiling, dev-time only	⚠️ Works, but no trustworthy number — two runs killed by our own bugs
NVIDIA NIM	Hosted VLM catalogue	❌ 4 of 5 models 404 — account entitlement, takes days
HuggingFace	Model weights, incl. gated Cosmos	✅
Kaggle	Free T4 fallback	Authenticated, never needed
Lightning AI	Considered	Not used — Modal covered it

Why Modal: one Python file, `@app.function(gpu="A100")`, jobs run headless and detached. Persistent volumes held the dataset so nothing re-uploaded. **I could drive it entirely from the terminal**, which is what made three agents on one GPU budget workable.

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6. The approach — and the two things that actually moved the score

+9.8 marks came in the final hour. Fourteen VLM interventions before that moved nothing.

The big one: span geometry, +6.1

Eight prediction sets — three model sizes, five window configs, a fine-tune, targeted prompts — **all scored exactly 8.0 on D3.**

That invariance was the clue. **If changing the model changes nothing, the failure isn't in the model.**

Our D3 spans were **1–8 seconds inside videos of 327–602 seconds**. A 5-second span can never reach 50% overlap with a 100-second event — *geometrically impossible, no matter how right the class is.*

We widened spans to 112–138s. **49.9** → **56.0.** No new inference. No changed class. **Our classes had been right all along.**

> The evidence was in our own notes from that morning: the practice data had a **125-second** congestion event. We wrote down that D3 events are long — then spent hours swapping models instead of spans.

The second: SigLIP2, +2.4

Frozen SigLIP2 → 8 frames → bidirectional GRU head → 12 classes. It predicts `loitering` on two videos — a class we score **0/6** on and **no VLM proposed all day**.

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7. What we'd say about the failures

Ten silent failures in one day. None crashed. Every one produced a plausible artifact.

Pick two or three of these to tell: - **ffmpeg's `-ss` is a keyframe seek** — silently returns *more* than you ask for. A 468s video rendered as 241s and every timestamp after it was wrong. - **The fix for one bug**

caused another. We added per-video error handling so one bad video couldn't kill a 34-video run. It then converted a loud crash into **10 rows of confident, plausible, wrong output.** - **SigLIP reported "validation accuracy 1.000".** Meaningless — 77 embeddings and a 7-sample validation split, because a 13GB upload had died unnoticed. - **We deleted 125 synthetic videos we had already rendered** because their timestamps were silently wrong. *We threw away a third of our training data rather than train on labels we couldn't verify.*

The line to land:

> **In a seven-hour build, every failure that mattered was silent. So we validated intermediate artifacts, not outputs.**

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8. Likely questions

"Why not just fine-tune?" We did. LoRA on Qwen3-VL-4B, 300 steps, loss converged. It scored **42.4** replacing the baseline and **49.2** added to it — both **below** the 49.9 zero-shot. It trained on 2,610 rows instead of the full set because an upload failed, and 71% of its targets were empty, teaching it to stay silent when our problem was recall.

"Is it real-time?" Measured, not asserted. **A window closes every 10 seconds and is scored in 1.83 seconds — 5.5× headroom, 2.8× at p95.** Self-reported latency, instrumented from the first call.

"What would you do next?" 1. **GRPO with a tIoU-shaped reward** — the literature says it beats SFT on temporal grounding, and 75 of the 100 marks are IoU-gated. 2. **Make SigLIP the cheap always-on stage** and the VLM the verifier. That's the cascade the brief actually asks for. 3. **Feed SigLIP the full dataset** — it earned +2.4 marks trained on a fifth of the data.

"What are you still bad at?" Four classes we never detect: `loitering` 0/6, `fighting` 0/3, `road_spill` 0/3, `stalled` 0/2. All **context and duration** classes. The VLM is confident on appearance and blind to anything defined by time rather than pixels.